

PORT COLBORNE, ON, Canada

WMO: 714630

Lat: 42.867N

Lon: 79.251W

Elev: 184

StdP: 99.14

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.2	-13.5	-21.0	0.6	-14.9	-18.6	0.7	-12.3	21.7	-2.6	20.0	-2.5	6.3	270	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	26.1	21.1	25.2	21.2	24.4	20.7	23.5	24.7	22.6	24.1	21.8	23.6	4.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	18.2	24.3	22.0	17.1	23.6	21.2	16.2	23.2	70.9	24.8	67.4	24.2	64.7	23.7	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	15.4	13.1	-18.3	28.8	3.5	1.6	-20.8	30.0	-22.9	30.9	-24.8	31.8	-27.4	33.0		
(5)			-18.8	24.0	3.5	1.1	-21.3	24.8	-23.3	25.4	-25.3	26.0	-27.8	26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-3.3	-3.2	-0.1	5.8	11.9	17.8	21.1	21.2	17.9	11.8	5.5	0.1
	DBStd	9.76	5.57	5.33	4.47	3.85	3.47	2.96	2.25	2.30	3.44	4.26	4.47	4.70
	HDD10.0	1731	412	369	314	133	19	0	0	0	1	31	146	307
	HDD18.3	3679	670	602	572	374	203	44	4	4	48	206	386	564
	CDD10.0	1342	0	0	0	8	77	234	343	346	237	86	10	1
	CDD18.3	250	0	0	0	0	3	28	89	92	35	3	0	0
	CDH23.3	481	0	0	0	1	7	39	156	212	66	1	0	0
	CDH26.7	21	0	0	0	0	1	3	8	6	3	0	0	0
Wind	WSAvg	6.0	7.7	6.7	5.8	5.6	5.2	4.8	5.0	5.1	5.4	6.6	7.3	7.4
Precipitation	PrecAvg	968	80	54	69	83	85	67	85	80	97	88	95	75
	PrecMax	1166	134	111	127	141	154	171	186	141	190	198	140	156
	PrecMin	795	40	29	36	41	20	18	27	30	37	35	43	31
	PrecStd	120	28	19	26	28	39	39	42	30	38	41	24	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.5	9.1	13.1	19.4	24.0	26.5	27.6	27.6	26.8	22.4	16.3	11.9
		MCWB	8.9	6.7	8.7	10.2	15.9	19.4	19.8	20.3	21.5	18.1	13.1	10.0
	2%	DB	7.5	6.8	9.7	15.9	20.9	24.4	25.7	26.1	24.9	20.8	14.7	9.5
		MCWB	6.1	4.7	6.6	9.2	14.8	19.0	20.8	21.8	21.4	17.6	12.2	7.7
	5%	DB	5.6	5.0	7.4	13.1	18.7	22.9	24.8	25.2	23.9	19.4	13.2	8.0
		MCWB	4.1	3.2	4.8	8.1	14.1	18.9	21.0	21.6	20.5	16.2	10.5	6.4
	10%	DB	4.0	3.2	5.6	11.0	16.8	21.7	24.1	24.5	22.8	17.8	11.9	6.2
		MCWB	2.5	1.6	3.4	7.1	13.1	18.5	20.6	21.1	19.6	14.9	9.2	4.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.3	7.5	9.6	13.0	18.4	22.6	24.2	25.0	23.3	19.6	14.2	10.4
		MCDB	10.1	8.5	11.8	16.9	22.1	24.0	24.9	25.5	24.7	21.3	15.5	12.1
	2%	WB	6.2	5.1	7.2	10.8	16.7	21.2	23.0	23.9	22.4	18.2	12.7	7.9
		MCDB	7.3	6.4	9.0	14.4	19.4	22.7	24.3	25.0	24.3	20.1	14.3	9.1
	5%	WB	4.3	3.2	5.3	9.1	15.2	20.4	22.3	22.9	21.3	16.8	10.8	6.4
		MCDB	5.4	4.6	7.2	11.9	18.1	22.1	23.8	24.4	23.3	18.8	12.8	7.8
	10%	WB	2.6	1.7	3.5	7.7	13.6	19.3	21.6	22.0	20.2	15.3	9.4	4.5
		MCDB	3.8	3.0	5.3	10.4	16.1	21.2	23.4	23.8	22.3	17.6	11.6	5.9
Mean Daily Temperature Range	5% DB	MDBR	5.9	6.0	5.9	6.7	6.1	5.0	4.5	5.0	5.6	5.5	5.3	4.9
		MCDBR	6.8	7.4	8.3	10.6	8.8	6.6	5.2	5.2	5.7	6.2	6.5	6.1
		MCWBR	6.5	6.3	6.0	6.5	5.0	3.9	3.8	4.0	3.9	4.6	5.8	5.9
	5% WB	MCDBR	6.9	7.2	7.9	9.5	7.6	5.4	4.1	4.3	5.2	5.8	6.0	6.2
		MCWBR	6.9	6.5	6.5	6.8	5.3	4.2	3.7	4.2	4.1	5.1	5.9	6.0
Clear-Sky Solar Irradiance	taub	0.301	0.335	0.335	0.376	0.405	0.434	0.437	0.425	0.382	0.362	0.315	0.301	
	taud	2.401	2.288	2.391	2.324	2.290	2.243	2.253	2.277	2.376	2.408	2.508	2.457	
	Ebn at Noon	836	854	907	890	870	842	834	833	846	818	819	804	
	Edh at Noon	78	103	105	122	129	136	134	126	106	91	70	68	
All-Sky Solar Radiation	RadAvg	1.32	2.12	3.26	4.41	5.41	5.87	5.91	5.24	4.15	2.52	1.56	1.04	
	RadStd	0.12	0.20	0.38	0.47	0.56	0.44	0.45	0.28	0.38	0.24	0.22	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page